

A Uniform-Derivative Proof of the Varying-Truncation Summability Condition

June 2026

Abstract

Moharrami, Murthy, Roy, and Srikant introduced a smooth truncation $\Lambda_\theta^{(M)}$ of the regenerative representation of the risk-sensitive exponential cost and assumed, in their varying-truncation policy-gradient theorem, that

$$\sum_{m=0}^{\infty} \sup_{\theta} (\Lambda_\theta^{(M_{m+1})} - \Lambda_\theta^{(M_m)}) < \infty.$$

Their numerical implementation suggests using the direct schedule $M_m = \gamma_m^{-\beta}$, and the question is whether this unrounded schedule satisfies the displayed summability condition. This note gives a solution of that summability problem, under the standing finite-state assumptions of their paper and for the same smooth truncation. In fact, a stronger statement holds: every deterministic eventually nondecreasing sequence $M_m \rightarrow \infty$ with $M_m > 1$ satisfies the above summability condition. The proof obtains the integrable uniform bound

$$0 \leq \partial_M \Lambda_\theta^{(M)} \leq AM^{-1-\varepsilon}$$

for all large M , uniformly in θ . Thus the direct schedule $M_m = \gamma_m^{-\beta}$ works whenever $\gamma_m \downarrow 0$ in the monotone sense. This solves only the truncation-summability subproblem; the separate martingale-noise condition in the policy-gradient theorem remains unchanged.

1 Introduction and status of the result

The purpose of this note is narrow. It addresses the condition on the *sequence of truncation levels* in the varying-truncation theorem of Moharrami–Murthy–Roy–Srikant [1]; it does not reprove the stochastic approximation theorem itself. The conclusion is therefore best stated as follows.

The argument below gives a solution, not merely partial progress, to the summability condition for the smooth truncations used in [1]. Once this condition is inserted into the authors' theorem, the remaining assumptions of that theorem, including the separate noise summability condition, must still be imposed.

The main point is that the particular fourth-order smooth truncation has a very small derivative with respect to the truncation level. A uniform $L^{1+\varepsilon}$ estimate near the true root turns this derivative into an integrable envelope in M . Summability then follows by integrating over the disjoint intervals $[M_m, M_{m+1}]$.

2 Setup

Let $\mathcal{X}$ be a finite state space and fix a distinguished regeneration state $x^* \in \mathcal{X}$. For $\theta \in \mathbb{R}^\ell$, let P_θ be a transition matrix and let $C_\theta : \mathcal{X} \rightarrow \mathbb{R}$ be the one-step cost. We use the standing finite-state hypotheses from [1]: the closure of the family of kernels consists of irreducible aperiodic kernels with the common recurrent state x^* ; the costs are uniformly bounded; and the regularity assumptions needed to define the regenerative risk-sensitive cost and its smooth truncations hold. Write

$$\bar{C} := \sup_{\theta, x} |C_\theta(x)| < \infty,$$

and let $\alpha > 0$ be the risk parameter. Starting from x^* , define

$$\tau := \inf\{n \geq 1 : \Phi_n = x^*\}$$

and, for $\Lambda \in \mathbb{R}$,

$$H_\theta(\Lambda) := \exp \left(\sum_{i=0}^{\tau-1} (\alpha C_\theta(\Phi_i) - \Lambda) \right).$$

The untruncated cost Λ_θ is characterized by the regenerative fixed point equation

$$g_\theta(\Lambda_\theta) = 1, \quad g_\theta(\Lambda) := \mathbb{E}_\theta[H_\theta(\Lambda) \mid \Phi_0 = x^*].$$

Equivalently, e^{Λ_θ} is the Perron–Frobenius eigenvalue of the nonnegative matrix with entries $e^{\alpha C_\theta(x)} P_\theta(x, y)$. Hence

$$|\Lambda_\theta| \leq \alpha \bar{C} \quad \text{for all } \theta, \tag{1}$$

because the row sums of that matrix lie between $e^{-\alpha \bar{C}}$ and $e^{\alpha \bar{C}}$.

For $M > 1$, define

$$P_4(t) := \sum_{j=0}^4 \frac{t^j}{j!}, \quad P_3(t) := \sum_{j=0}^3 \frac{t^j}{j!},$$

and define the smooth truncation

$$\phi_M(z) := z \mathbf{1}_{\{z \leq M\}} + M P_4 \left(\log \frac{z}{M} \right) \mathbf{1}_{\{z > M\}}, \quad z > 0. \tag{2}$$

The truncated fixed-point function and its root are

$$g_\theta^{(M)}(\Lambda) := \mathbb{E}_\theta[\phi_M(H_\theta(\Lambda)) \mid \Phi_0 = x^*], \quad g_\theta^{(M)}(\Lambda_\theta^{(M)}) = 1.$$

The root is unique because $g_\theta^{(M)}$ is continuous, strictly decreasing, tends to infinity as $\Lambda \rightarrow -\infty$, and tends to zero as $\Lambda \rightarrow \infty$.

Two elementary monotonicity facts will be used repeatedly. First, $P_4(t) \leq e^t$ for $t \geq 0$, so

$$0 < \phi_M(z) \leq z, \quad z > 0. \tag{3}$$

Thus $g_\theta^{(M)}(\Lambda_\theta) \leq 1$, and since $g_\theta^{(M)}$ is decreasing,

$$\Lambda_\theta^{(M)} \leq \Lambda_\theta. \tag{4}$$

Second, for fixed z , $M \mapsto \phi_M(z)$ is nondecreasing. Hence $M \mapsto \Lambda_\theta^{(M)}$ is nondecreasing.

3 Uniform recurrence and a uniform moment margin

The proof needs a uniform moment strictly beyond order one for $H_\theta(\Lambda)$ when Λ lies slightly below the true root Λ_θ . The following recurrence lemma is the finite-state input.

Lemma 1 (Uniform exponential cycle tails). *There exist constants $a_0 > 0$ and $K_0 < \infty$ such that*

$$\sup_{\theta} \sup_{x \in \mathcal{X}} \mathbb{E}_{\theta}^x [e^{a_0 \sigma_{x^*}}] \leq K_0, \quad \sigma_{x^*} := \inf\{n \geq 0 : \Phi_n = x^*\}.$$

Consequently, when the chain starts from x^ and $\tau := \inf\{n \geq 1 : \Phi_n = x^*\}$ denotes the first positive return time, then*

$$\sup_{\theta} \mathbb{E}_{\theta}^{x^*} [e^{a'_0 \tau}] \leq K'_0$$

for some $a'_0 > 0$ and $K'_0 < \infty$. The analogous bounds also hold uniformly for the twisted kernels associated with $(P_\theta, C_\theta, \Lambda_\theta)$.

Proof. Let $\mathcal{K}$ be the compact closure of $\{P_\theta : \theta \in \mathbb{R}^\ell\}$. Let $d = |\mathcal{X}|$. For every $P \in \mathcal{K}$ and every $x \in \mathcal{X}$, irreducibility implies that either $x = x^*$ or there is a simple path from x to x^* of length at most $d - 1$ and positive P -probability. Thus

$$\mathbb{P}_P^x(\sigma_{x^*} \leq d - 1) > 0$$

for every P and x , with the probability equal to one when $x = x^*$. Since only finitely many paths of length at most $d - 1$ exist, the map

$$P \mapsto \min_{x \in \mathcal{X}} \mathbb{P}_P^x(\sigma_{x^*} \leq d - 1)$$

is a minimum of finitely many maxima of continuous monomials in the entries of P . It is positive on the compact set $\mathcal{K}$, so its minimum is some $p > 0$.

The Markov property applied at times $d - 1, 2(d - 1), \dots$ then gives a uniform geometric tail for σ_{x^*} ; increasing the block length to one if $d = 1$ is harmless. This proves a uniform exponential moment for σ_{x^*} .

It remains to pass from hitting times to the first positive return time when the initial state is x^* . After one transition from x^* , the chain is at some state Y . The remaining time to hit x^* is stochastically controlled, uniformly in Y , by the preceding hitting-time estimate. Hence τ has a uniform exponential moment, possibly with a smaller exponent.

For the twisted kernels, let h_θ be the positive right Perron–Frobenius eigenvector, normalized by $h_\theta(x^*) = 1$, of the matrix $F_\theta(x, y) := e^{\alpha C_\theta(x)} P_\theta(x, y)$. The twisted transition matrix is

$$\tilde{P}_\theta(x, y) := e^{\alpha C_\theta(x) - \Lambda_\theta} P_\theta(x, y) \frac{h_\theta(y)}{h_\theta(x)}.$$

By finite-dimensional compactness and Perron–Frobenius continuity, the closure of $\{\tilde{P}_\theta\}$ is again a compact family of irreducible kernels on $\mathcal{X}$. Repeating the preceding hitting-time and return-time argument for this family proves the same type of uniform exponential-tail estimate. $\square$

Lemma 2 (Uniform $L^{1+\varepsilon}$ room near the true root). *There exist constants $\delta > 0$, $\varepsilon > 0$, and $K < \infty$ such that, for all θ and all Λ satisfying $0 \leq \Lambda_\theta - \Lambda \leq \delta$,*

$$\mathbb{E}_\theta[(1 + \tau)H_\theta(\Lambda)^{1+\varepsilon} \mid \Phi_0 = x^*] \leq K. \quad (5)$$

Proof. Let $\tilde{\mathbb{E}}_\theta$ denote expectation for the twisted kernel. Since the cycle starts and ends at x^* and $h_\theta(x^*) = 1$, the random variable $H_\theta(\Lambda_\theta)$ is the Radon–Nikodym derivative of the twisted cycle law with respect to the original cycle law. Therefore

$$\mathbb{E}_\theta[(1 + \tau)H_\theta(\Lambda)^{1+\varepsilon}] = \tilde{\mathbb{E}}_\theta \left[(1 + \tau) \exp \left((\Lambda_\theta - \Lambda)\tau + \varepsilon \sum_{i=0}^{\tau-1} (\alpha C_\theta(\Phi_i) - \Lambda) \right) \right].$$

By (1), if $0 \leq \Lambda_\theta - \Lambda \leq \delta$, then $|\alpha C_\theta(x) - \Lambda| \leq 2\alpha\bar{C} + \delta$ for all x . Choose $\delta > 0$ and then $\varepsilon > 0$ so small that

$$\delta + \varepsilon(2\alpha\bar{C} + \delta) < a'_0,$$

where a'_0 is a uniform exponential-moment exponent for the twisted positive return time. The preceding display is then bounded above by

$$\tilde{\mathbb{E}}_\theta[(1 + \tau)e^{b\tau}], \quad b := \delta + \varepsilon(2\alpha\bar{C} + \delta) < a'_0.$$

The uniform exponential moment at rate a'_0 for the twisted kernels implies a uniform bound on this last quantity. This proves (5). $\square$

4 Localization of the truncated root

Lemma 3 (Eventual uniform localization). *With δ as in Lemma 2, there exists $M_0 > 1$ such that, for all $M \geq M_0$ and all θ ,*

$$0 \leq \Lambda_\theta - \Lambda_\theta^{(M)} \leq \delta/2. \quad (6)$$

Proof. The left inequality is (4). For the right inequality, put $\Lambda = \Lambda_\theta - \delta/2$. The twisted-kernel identity gives

$$g_\theta(\Lambda_\theta - \delta/2) = \tilde{\mathbb{E}}_\theta[e^{(\delta/2)\tau}] \geq e^{\delta/2},$$

because $\tau \geq 1$. On the other hand, by (3),

$$0 \leq g_\theta(\Lambda) - g_\theta^{(M)}(\Lambda) \leq \mathbb{E}_\theta[H_\theta(\Lambda)\mathbf{1}_{\{H_\theta(\Lambda) > M\}}].$$

Lemma 2 and Markov's inequality bound the last term by $KM^{-\varepsilon}$, uniformly in θ . Choose M_0 so large that $KM_0^{-\varepsilon} < e^{\delta/2} - 1$. Then $g_\theta^{(M)}(\Lambda_\theta - \delta/2) > 1$ for all $M \geq M_0$. Since $g_\theta^{(M)}$ is decreasing and equals one at $\Lambda_\theta^{(M)}$, this implies $\Lambda_\theta^{(M)} > \Lambda_\theta - \delta/2$. $\square$

5 The uniform integrable derivative

The decisive calculation is the derivative of (2) with respect to M . For $z > M$ and $t = \log(z/M)$,

$$\frac{\partial}{\partial M}(MP_4(t)) = P_4(t) - P_3(t) = \frac{t^4}{4!}.$$

Thus, for fixed $z > 0$,

$$\partial_M \phi_M(z) = \frac{1}{4!} \left(\log \frac{z}{M} \right)^4 \mathbf{1}_{\{z > M\}}. \quad (7)$$

Similarly, if $z = H_\theta(\Lambda)$, then

$$-\partial_\Lambda \phi_M(H_\theta(\Lambda)) = \tau \left[H_\theta(\Lambda) \mathbf{1}_{\{H_\theta(\Lambda) \leq M\}} + MP_3 \left(\log \frac{H_\theta(\Lambda)}{M} \right) \mathbf{1}_{\{H_\theta(\Lambda) > M\}} \right]. \quad (8)$$

The derivatives are continuous across the boundary $z = M$. The domination provided by Lemma 2, after the localization of Lemma 3, justifies differentiating the expectations in a neighborhood of the root: for $u \geq 1$ and any $q > 0$, polynomial-logarithmic terms such as $P_4(\log u)$, $P_3(\log u)$, and $(\log u)^4$ are bounded by constants times u^q , while the factor τ in (8) is covered by (5). Hence the implicit function theorem applies.

Lemma 4 (Uniform integrable derivative). *There are constants $M_1 \geq M_0$ and $A < \infty$ such that, for all $M \geq M_1$ and all θ ,*

$$0 \leq \frac{\partial}{\partial M} \Lambda_\theta^{(M)} \leq AM^{-1-\varepsilon}, \quad (9)$$

where ε is the exponent from Lemma 2.

Proof. Fix $M \geq M_0$ and write $H := H_\theta(\Lambda_\theta^{(M)})$. By Lemmas 2 and 3,

$$\sup_{\theta, M \geq M_0} \mathbb{E}_\theta[H^{1+\varepsilon}] \leq K. \quad (10)$$

Let $q := 1 + \varepsilon$. There is a finite constant B_q such that, for all $u \geq 1$,

$$P_4(\log u) \leq B_q u^q, \quad \frac{(\log u)^4}{4!} \leq B_q u^q. \quad (11)$$

The tail contribution to the fixed-point identity satisfies

$$\mathbb{E}_\theta[MP_4(\log(H/M))\mathbf{1}_{\{H > M\}}] \leq B_q M^{1-q} \mathbb{E}_\theta[H^q] \leq B_q K M^{-\varepsilon}.$$

Choose $M_1 \geq M_0$ so large that $B_q K M_1^{-\varepsilon} \leq 1/2$. Since $\mathbb{E}_\theta[\phi_M(H)] = 1$, it follows that, for $M \geq M_1$,

$$\mathbb{E}_\theta[H\mathbf{1}_{\{H \leq M\}}] = 1 - \mathbb{E}_\theta[MP_4(\log(H/M))\mathbf{1}_{\{H > M\}}] \geq \frac{1}{2}. \quad (12)$$

Using (8), $\tau \geq 1$, and positivity of P_3 on $[0, \infty)$, we get

$$-\partial_\Lambda g_\theta^{(M)}(\Lambda_\theta^{(M)}) \geq \mathbb{E}_\theta[H\mathbf{1}_{\{H \leq M\}}] \geq \frac{1}{2}. \quad (13)$$

By (7), (11), and (10),

$$\begin{aligned} \partial_M g_\theta^{(M)}(\Lambda_\theta^{(M)}) &= \mathbb{E}_\theta \left[\frac{1}{4!} \left(\log \frac{H}{M} \right)^4 \mathbf{1}_{\{H > M\}} \right] \\ &\leq B_q M^{-q} \mathbb{E}_\theta[H^q] \leq B_q K M^{-1-\varepsilon}. \end{aligned} \quad (14)$$

The implicit derivative formula is

$$\partial_M \Lambda_\theta^{(M)} = \frac{\partial_M g_\theta^{(M)}(\Lambda_\theta^{(M)})}{-\partial_\Lambda g_\theta^{(M)}(\Lambda_\theta^{(M)})}.$$

Combining (13) and (14) gives (9) with $A = 2B_q K$. Nonnegativity follows from monotonicity of $M \mapsto \phi_M$ and therefore of $M \mapsto \Lambda_\theta^{(M)}$. $\square$

6 Summability for arbitrary monotone truncation schedules

Theorem 1 (The varying-truncation summability condition). *Let $(M_m)_{m \geq 0}$ be a deterministic sequence with $M_m > 1$, $M_m \rightarrow \infty$, and M_m eventually nondecreasing. Then*

$$\sum_{m=0}^{\infty} \sup_{\theta \in \mathbb{R}^\ell} (\Lambda_\theta^{(M_{m+1})} - \Lambda_\theta^{(M_m)}) < \infty. \quad (15)$$

In particular, if the sequence is nondecreasing from the start, it satisfies the varying-truncation summability assumption of [1].

Proof. Changing or deleting finitely many initial terms cannot affect convergence of the series, provided those terms are finite. We first note the required uniform finiteness on bounded M -intervals. If $M \geq M_{\min} > 1$, then (4) and (1) give $\Lambda_\theta^{(M)} \leq \alpha \bar{C}$. Conversely, set

$$L := -\alpha \bar{C} - \log M_{\min}.$$

For every cycle, each term $\alpha C_\theta(\Phi_i) - L$ is at least $\log M_{\min}$, so $H_\theta(L) \geq M_{\min}$. Since $M \geq M_{\min}$, if $H_\theta(L) \leq M$ then $\phi_M(H_\theta(L)) = H_\theta(L) \geq M_{\min}$, while if $H_\theta(L) > M$ then $\phi_M(H_\theta(L)) \geq M \geq M_{\min}$. Therefore $g_\theta^{(M)}(L) \geq M_{\min} > 1$. The root relation implies $\Lambda_\theta^{(M)} \geq L$. Hence $\Lambda_\theta^{(M)}$ is uniformly bounded when M ranges over any interval $[M_{\min}, M_1]$.

It remains to control the tail of the series. Choose m_0 so large that the sequence is nondecreasing for $m \geq m_0$ and $M_m \geq M_1$ for all $m \geq m_0$. By Lemma 4, for any $M' \geq M \geq M_1$ and any θ ,

$$0 \leq \Lambda_\theta^{(M')} - \Lambda_\theta^{(M)} = \int_M^{M'} \partial_u \Lambda_\theta^{(u)} \, du \leq A \int_M^{M'} u^{-1-\varepsilon} \, du.$$

Therefore

$$\begin{aligned} \sum_{m=m_0}^{\infty} \sup_{\theta} (\Lambda_\theta^{(M_{m+1})} - \Lambda_\theta^{(M_m)}) &\leq A \sum_{m=m_0}^{\infty} \int_{M_m}^{M_{m+1}} u^{-1-\varepsilon} \, du \\ &\leq A \int_{M_{m_0}}^{\infty} u^{-1-\varepsilon} \, du = \frac{A}{\varepsilon} M_{m_0}^{-\varepsilon} < \infty. \end{aligned}$$

Together with the finite initial segment, this proves (15). $\square$

Corollary 1 (The direct $\gamma^{-\beta}$ schedule). *Let $\gamma_m \downarrow 0$ in the monotone sense and fix $\beta > 0$. Put $M_m := \gamma_m^{-\beta}$. Then there is an index m_0 such that*

$$\sum_{m=m_0}^{\infty} \sup_{\theta \in \mathbb{R}^\ell} (\Lambda_\theta^{(\gamma_{m+1}^{-\beta})} - \Lambda_\theta^{(\gamma_m^{-\beta})}) < \infty.$$

Equivalently, after the harmless modification or deletion of finitely many initial truncation levels so that all displayed truncations are in the admissible range $M > 1$, the full summability condition holds. Consequently the unrounded truncation levels used in the numerical schedule satisfy the varying-truncation summability condition. The separate noise condition appearing in the policy-gradient theorem remains

$$\sum_m \gamma_m^2 M_m^2 < \infty,$$

which for $M_m = \gamma_m^{-\beta}$ becomes

$$\sum_m \gamma_m^{2(1-\beta)} < \infty.$$

Proof. Since $\gamma_m \downarrow 0$, the sequence $M_m = \gamma_m^{-\beta}$ is nondecreasing and tends to infinity. Also $M_m > 1$ for all sufficiently large m . Apply Theorem 1 to the tail. Finite initial changes do not affect convergence of the series. The displayed form of the noise condition follows from the identity $\gamma_m^2 M_m^2 = \gamma_m^{2(1-\beta)}$. $\square$

7 Concluding remarks

A crude uniform approximation estimate of the form $\sup_{\theta}(\Lambda_{\theta} - \Lambda_{\theta}^{(M)}) = O(M^{-\varepsilon})$ would not by itself imply (15) for arbitrary slowly increasing M_m , because $\sum_m M_m^{-\varepsilon}$ may diverge. The stronger and more useful fact is the integrable derivative estimate $\partial_M \Lambda_{\theta}^{(M)} = O(M^{-1-\varepsilon})$, uniformly in θ . This converts the summability problem into the integral of an integrable envelope over disjoint truncation intervals.

Thus, subject to the same finite-state assumptions and the same smooth truncation used in [1], the note obtains a solution of the truncation-summability problem. It should not be read as an independent proof of the full policy-gradient convergence theorem; rather, it verifies the specific truncation schedule condition needed to invoke that theorem.

References

- [1] M. Moharrami, Y. Murthy, A. Roy, and R. Srikant, A policy gradient algorithm for the risk-sensitive exponential cost MDP, *Mathematics of Operations Research* 50(1):431–458, 2025. DOI: 10.1287/moor.2022.0139. Also available as arXiv:2202.04157.